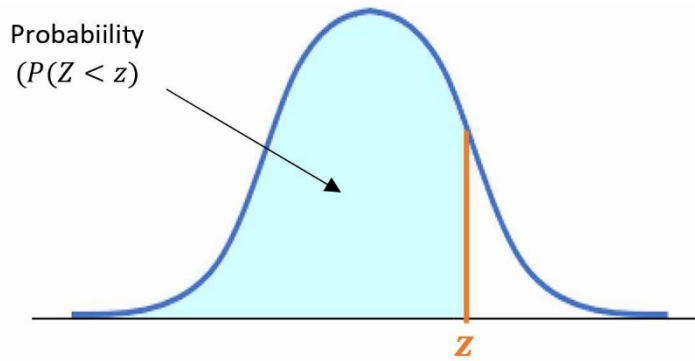
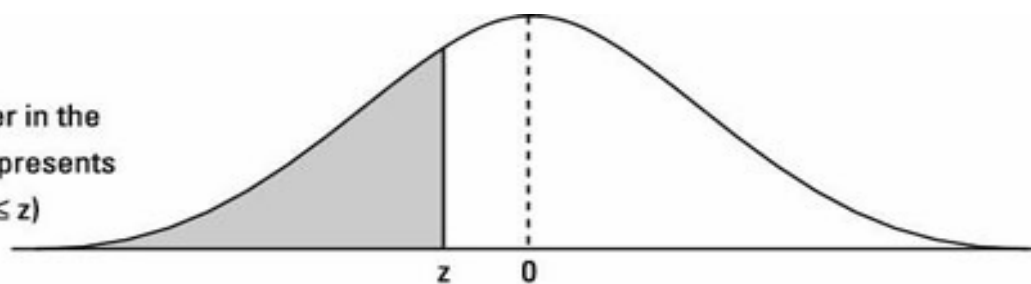


## Normal distribution tables

[illegible]

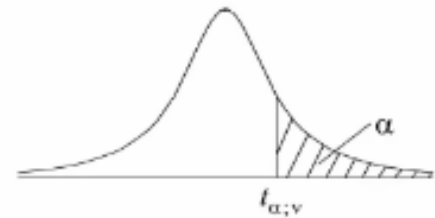
Number in the  
table represents  
 $P(Z \leq z)$



| $z$  | 0.00  | 0.01  | 0.02  | 0.03  | 0.04  | 0.05  | 0.06  | 0.07  | 0.08  | 0.09  |
|------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| -3.6 | .0002 | .0002 | .0001 | .0001 | .0001 | .0001 | .0001 | .0001 | .0001 | .0001 |
| -3.5 | .0002 | .0002 | .0002 | .0002 | .0002 | .0002 | .0002 | .0002 | .0002 | .0002 |
| -3.4 | .0003 | .0003 | .0003 | .0003 | .0003 | .0003 | .0003 | .0003 | .0003 | .0002 |
| -3.3 | .0005 | .0005 | .0005 | .0004 | .0004 | .0004 | .0004 | .0004 | .0004 | .0003 |
| -3.2 | .0007 | .0007 | .0006 | .0006 | .0006 | .0006 | .0006 | .0005 | .0005 | .0005 |
| -3.1 | .0010 | .0009 | .0009 | .0009 | .0008 | .0008 | .0008 | .0008 | .0007 | .0007 |
| -3.0 | .0013 | .0013 | .0013 | .0012 | .0012 | .0011 | .0011 | .0011 | .0010 | .0010 |
| -2.9 | .0019 | .0018 | .0018 | .0017 | .0016 | .0016 | .0015 | .0015 | .0014 | .0014 |
| -2.8 | .0026 | .0025 | .0024 | .0023 | .0023 | .0022 | .0021 | .0021 | .0020 | .0019 |
| -2.7 | .0035 | .0034 | .0033 | .0032 | .0031 | .0030 | .0029 | .0028 | .0027 | .0026 |
| -2.6 | .0047 | .0045 | .0044 | .0043 | .0041 | .0040 | .0039 | .0038 | .0037 | .0036 |
| -2.5 | .0062 | .0060 | .0059 | .0057 | .0055 | .0054 | .0052 | .0051 | .0049 | .0048 |
| -2.4 | .0082 | .0080 | .0078 | .0075 | .0073 | .0071 | .0069 | .0068 | .0066 | .0064 |
| -2.3 | .0107 | .0104 | .0102 | .0099 | .0096 | .0094 | .0091 | .0089 | .0087 | .0084 |
| -2.2 | .0139 | .0136 | .0132 | .0129 | .0125 | .0122 | .0119 | .0116 | .0113 | .0110 |
| -2.1 | .0179 | .0174 | .0170 | .0166 | .0162 | .0158 | .0154 | .0150 | .0146 | .0143 |
| -2.0 | .0228 | .0222 | .0217 | .0212 | .0207 | .0202 | .0197 | .0192 | .0188 | .0183 |
| -1.9 | .0287 | .0281 | .0274 | .0268 | .0262 | .0256 | .0250 | .0244 | .0239 | .0233 |
| -1.8 | .0359 | .0351 | .0344 | .0336 | .0329 | .0322 | .0314 | .0307 | .0301 | .0294 |
| -1.7 | .0446 | .0436 | .0427 | .0418 | .0409 | .0401 | .0392 | .0384 | .0375 | .0367 |
| -1.6 | .0548 | .0537 | .0526 | .0516 | .0505 | .0495 | .0485 | .0475 | .0465 | .0455 |
| -1.5 | .0668 | .0655 | .0643 | .0630 | .0618 | .0606 | .0594 | .0582 | .0571 | .0559 |
| -1.4 | .0808 | .0793 | .0778 | .0764 | .0749 | .0735 | .0721 | .0708 | .0694 | .0681 |
| -1.3 | .0968 | .0951 | .0934 | .0918 | .0901 | .0885 | .0869 | .0853 | .0838 | .0823 |
| -1.2 | .1151 | .1131 | .1112 | .1093 | .1075 | .1056 | .1038 | .1020 | .1003 | .0985 |
| -1.1 | .1357 | .1335 | .1314 | .1292 | .1271 | .1251 | .1230 | .1210 | .1190 | .1170 |
| -1.0 | .1587 | .1562 | .1539 | .1515 | .1492 | .1469 | .1446 | .1423 | .1401 | .1379 |
| -0.9 | .1841 | .1814 | .1788 | .1762 | .1736 | .1711 | .1685 | .1660 | .1635 | .1611 |
| -0.8 | .2119 | .2090 | .2061 | .2033 | .2005 | .1977 | .1949 | .1922 | .1894 | .1867 |
| -0.7 | .2420 | .2389 | .2358 | .2327 | .2296 | .2266 | .2236 | .2206 | .2177 | .2148 |
| -0.6 | .2743 | .2709 | .2676 | .2643 | .2611 | .2578 | .2546 | .2514 | .2483 | .2451 |
| -0.5 | .3085 | .3050 | .3015 | .2981 | .2946 | .2912 | .2877 | .2843 | .2810 | .2776 |
| -0.4 | .3446 | .3409 | .3372 | .3336 | .3300 | .3264 | .3228 | .3192 | .3156 | .3121 |
| -0.3 | .3821 | .3783 | .3745 | .3707 | .3669 | .3632 | .3594 | .3557 | .3520 | .3483 |
| -0.2 | .4207 | .4168 | .4129 | .4090 | .4052 | .4013 | .3974 | .3936 | .3897 | .3859 |
| -0.1 | .4602 | .4562 | .4522 | .4483 | .4443 | .4404 | .4364 | .4325 | .4286 | .4247 |
| -0.0 | .5000 | .4960 | .4920 | .4880 | .4840 | .4801 | .4761 | .4721 | .4681 | .4641 |

## Table of the Student's $t$ -distribution

The table gives the values of  $t_{\alpha;v}$  where  
 $\Pr(T_v > t_{\alpha;v}) = \alpha$ , with  $v$  degrees of freedom



| $\alpha \backslash v$ | 0.1   | 0.05  | 0.025  | 0.01   | 0.005  | 0.001   | 0.0005  |
|-----------------------|-------|-------|--------|--------|--------|---------|---------|
| 1                     | 3.078 | 6.314 | 12.076 | 31.821 | 63.657 | 318.310 | 636.620 |
| 2                     | 1.886 | 2.920 | 4.303  | 6.965  | 9.925  | 22.326  | 31.598  |
| 3                     | 1.638 | 2.353 | 3.182  | 4.541  | 5.841  | 10.213  | 12.924  |
| 4                     | 1.533 | 2.132 | 2.776  | 3.747  | 4.604  | 7.173   | 8.610   |
| 5                     | 1.476 | 2.015 | 2.571  | 3.365  | 4.032  | 5.893   | 6.869   |
| 6                     | 1.440 | 1.943 | 2.447  | 3.143  | 3.707  | 5.208   | 5.959   |
| 7                     | 1.415 | 1.895 | 2.365  | 2.998  | 3.499  | 4.785   | 5.408   |
| 8                     | 1.397 | 1.860 | 2.306  | 2.896  | 3.355  | 4.501   | 5.041   |
| 9                     | 1.383 | 1.833 | 2.262  | 2.821  | 3.250  | 4.297   | 4.781   |
| 10                    | 1.372 | 1.812 | 2.228  | 2.764  | 3.169  | 4.144   | 4.587   |
| 11                    | 1.363 | 1.796 | 2.201  | 2.718  | 3.106  | 4.025   | 4.437   |
| 12                    | 1.356 | 1.782 | 2.179  | 2.681  | 3.055  | 3.930   | 4.318   |
| 13                    | 1.350 | 1.771 | 2.160  | 2.650  | 3.012  | 3.852   | 4.221   |
| 14                    | 1.345 | 1.761 | 2.145  | 2.624  | 2.977  | 3.787   | 4.140   |
| 15                    | 1.341 | 1.753 | 2.131  | 2.602  | 2.947  | 3.733   | 4.073   |
| 16                    | 1.337 | 1.746 | 2.120  | 2.583  | 2.921  | 3.686   | 4.015   |
| 17                    | 1.333 | 1.740 | 2.110  | 2.567  | 2.898  | 3.646   | 3.965   |
| 18                    | 1.330 | 1.734 | 2.101  | 2.552  | 2.878  | 3.610   | 3.922   |
| 19                    | 1.328 | 1.729 | 2.093  | 2.539  | 2.861  | 3.579   | 3.883   |
| 20                    | 1.325 | 1.725 | 2.086  | 2.528  | 2.845  | 3.552   | 3.850   |
| 21                    | 1.323 | 1.721 | 2.080  | 2.518  | 2.831  | 3.527   | 3.819   |
| 22                    | 1.321 | 1.717 | 2.074  | 2.508  | 2.819  | 3.505   | 3.792   |
| 23                    | 1.319 | 1.714 | 2.069  | 2.500  | 2.807  | 3.485   | 3.767   |
| 24                    | 1.318 | 1.711 | 2.064  | 2.492  | 2.797  | 3.467   | 3.745   |
| 25                    | 1.316 | 1.708 | 2.060  | 2.485  | 2.787  | 3.450   | 3.725   |
| 26                    | 1.315 | 1.706 | 2.056  | 2.479  | 2.779  | 3.435   | 3.707   |
| 27                    | 1.314 | 1.703 | 2.052  | 2.473  | 2.771  | 3.421   | 3.690   |
| 28                    | 1.313 | 1.701 | 2.048  | 2.467  | 2.763  | 3.408   | 3.674   |
| 29                    | 1.311 | 1.699 | 2.045  | 2.462  | 2.756  | 3.396   | 3.659   |
| 30                    | 1.310 | 1.697 | 2.042  | 2.457  | 2.750  | 3.385   | 3.646   |
| 40                    | 1.303 | 1.684 | 2.021  | 2.423  | 2.704  | 3.307   | 3.551   |
| 60                    | 1.296 | 1.671 | 2.000  | 2.390  | 2.660  | 3.232   | 3.460   |
| 120                   | 1.289 | 1.658 | 1.980  | 2.358  | 2.617  | 3.160   | 3.373   |
| $\infty$              | 1.282 | 1.645 | 1.960  | 2.326  | 2.576  | 3.090   | 3.291   |

### Percentage Points of the Chi-Square Distribution

| Degrees of Freedom | Probability of a larger value of $\chi^2$ |        |        |        |        |       |       |       |       |
|--------------------|-------------------------------------------|--------|--------|--------|--------|-------|-------|-------|-------|
|                    | 0.99                                      | 0.95   | 0.90   | 0.75   | 0.50   | 0.25  | 0.10  | 0.05  | 0.01  |
| 1                  | 0.000                                     | 0.004  | 0.016  | 0.102  | 0.455  | 1.32  | 2.71  | 3.84  | 6.63  |
| 2                  | 0.020                                     | 0.103  | 0.211  | 0.575  | 1.386  | 2.77  | 4.61  | 5.99  | 9.21  |
| 3                  | 0.115                                     | 0.352  | 0.584  | 1.212  | 2.366  | 4.11  | 6.25  | 7.81  | 11.34 |
| 4                  | 0.297                                     | 0.711  | 1.064  | 1.923  | 3.357  | 5.39  | 7.78  | 9.49  | 13.28 |
| 5                  | 0.554                                     | 1.145  | 1.610  | 2.675  | 4.351  | 6.63  | 9.24  | 11.07 | 15.09 |
| 6                  | 0.872                                     | 1.635  | 2.204  | 3.455  | 5.348  | 7.84  | 10.64 | 12.59 | 16.81 |
| 7                  | 1.239                                     | 2.167  | 2.833  | 4.255  | 6.346  | 9.04  | 12.02 | 14.07 | 18.48 |
| 8                  | 1.647                                     | 2.733  | 3.490  | 5.071  | 7.344  | 10.22 | 13.36 | 15.51 | 20.09 |
| 9                  | 2.088                                     | 3.325  | 4.168  | 5.899  | 8.343  | 11.39 | 14.68 | 16.92 | 21.67 |
| 10                 | 2.558                                     | 3.940  | 4.865  | 6.737  | 9.342  | 12.55 | 15.99 | 18.31 | 23.21 |
| 11                 | 3.053                                     | 4.575  | 5.578  | 7.584  | 10.341 | 13.70 | 17.28 | 19.68 | 24.72 |
| 12                 | 3.571                                     | 5.226  | 6.304  | 8.438  | 11.340 | 14.85 | 18.55 | 21.03 | 26.22 |
| 13                 | 4.107                                     | 5.892  | 7.042  | 9.299  | 12.340 | 15.98 | 19.81 | 22.36 | 27.69 |
| 14                 | 4.660                                     | 6.571  | 7.790  | 10.165 | 13.339 | 17.12 | 21.06 | 23.68 | 29.14 |
| 15                 | 5.229                                     | 7.261  | 8.547  | 11.037 | 14.339 | 18.25 | 22.31 | 25.00 | 30.58 |
| 16                 | 5.812                                     | 7.962  | 9.312  | 11.912 | 15.338 | 19.37 | 23.54 | 26.30 | 32.00 |
| 17                 | 6.408                                     | 8.672  | 10.085 | 12.792 | 16.338 | 20.49 | 24.77 | 27.59 | 33.41 |
| 18                 | 7.015                                     | 9.390  | 10.865 | 13.675 | 17.338 | 21.60 | 25.99 | 28.87 | 34.80 |
| 19                 | 7.633                                     | 10.117 | 11.651 | 14.562 | 18.338 | 22.72 | 27.20 | 30.14 | 36.19 |
| 20                 | 8.260                                     | 10.851 | 12.443 | 15.452 | 19.337 | 23.83 | 28.41 | 31.41 | 37.57 |
| 22                 | 9.542                                     | 12.338 | 14.041 | 17.240 | 21.337 | 26.04 | 30.81 | 33.92 | 40.29 |
| 24                 | 10.856                                    | 13.848 | 15.659 | 19.037 | 23.337 | 28.24 | 33.20 | 36.42 | 42.98 |
| 26                 | 12.198                                    | 15.379 | 17.292 | 20.843 | 25.336 | 30.43 | 35.56 | 38.89 | 45.64 |
| 28                 | 13.565                                    | 16.928 | 18.939 | 22.657 | 27.336 | 32.62 | 37.92 | 41.34 | 48.28 |
| 30                 | 14.953                                    | 18.493 | 20.599 | 24.478 | 29.336 | 34.80 | 40.26 | 43.77 | 50.89 |
| 40                 | 22.164                                    | 26.509 | 29.051 | 33.660 | 39.335 | 45.62 | 51.80 | 55.76 | 63.69 |
| 50                 | 27.707                                    | 34.764 | 37.689 | 42.942 | 49.335 | 56.33 | 63.17 | 67.50 | 76.15 |
| 60                 | 37.485                                    | 43.188 | 46.459 | 52.294 | 59.335 | 66.98 | 74.40 | 79.08 | 88.38 |